

830,000 m² of green land built over?

*Testing a newsroom's satellite estimate of land lost to development in Malta
against an independent land-cover model*

THE CLAIM UNDER REVIEW

***“Nearly 830,000 square metres of nature
and cropland have been built up in Malta
between 2018 and 2023.”***

Amphora Media, Green to Grey, 11 September 2026.

Companion piece: nearly 95% of the take-up was farmland.

VERDICT: LARGELY SUPPORTED

Figure stands up, probably low; the 95% farmland share is not shown

TL;DR

In September 2026 Amphora Media, an investigative newsroom, reported that “nearly 830,000 square metres of nature and cropland have been built up in Malta between 2018 and 2023”, and that in “nearly 95% of the take-up, farmland rather than natural area was affected”. The press is held to the same standard as everyone else here. We checked the arithmetic and repeated the measurement with a different satellite land-cover product.

- 1

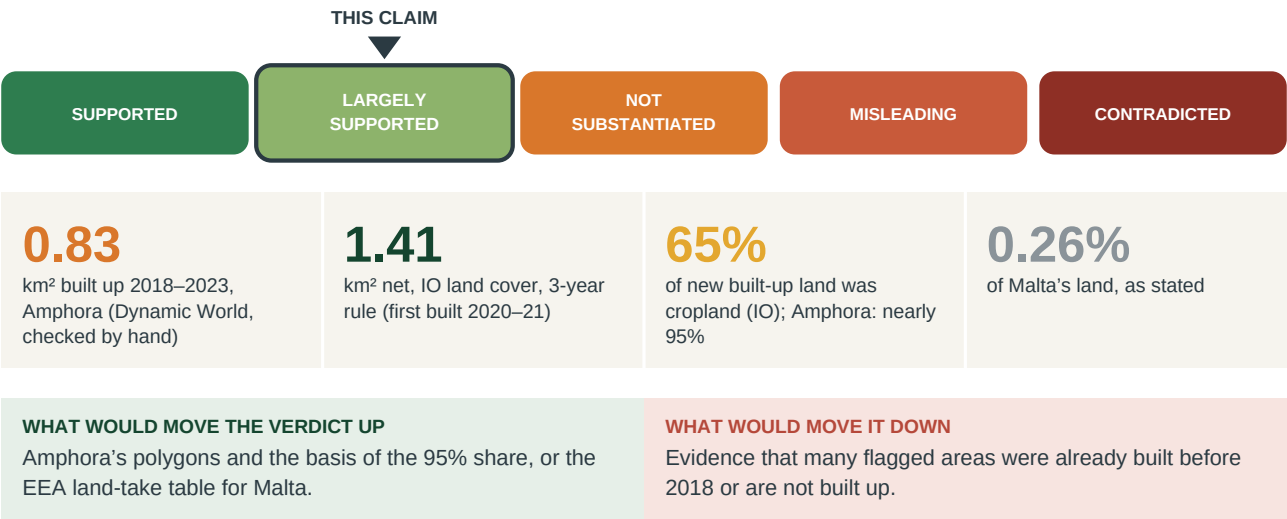
The arithmetic holds. 830,000 m² is 0.26% of Malta’s land and 116 football pitches, as stated, and a quarter to 0.3 of Comino, depending on the island area used (Amphora: “roughly a quarter”). The EEA’s own chart puts Malta’s 2012–2018 land take at about 0.91–0.94 km², the highest share of any of 39 European countries, matching the 920,000 m² quoted.
- 2

An independent measurement finds more, not less. Using Impact Observatory’s 10 m land-cover maps and counting only land that changed consistently, about 1.4 km² net was first mapped as built up in 2020 or 2021, a shorter window than Amphora’s 2018–2023. The model and method differ, so this is not like for like, but Amphora’s figure is lower, consistent with its own description as “a conservative estimate”.
- 3

The 95% farmland share was not reproduced. In the independent data, 65% of the new built-up land was cropland and 31% shrub or grassland. Abandoned fields can fall in either class, and Amphora does not say how it measured the 95% or for which period.
- 4

Some parts could not be checked. Amphora’s map polygons were not downloaded, the 2006–2012 EEA figure was not seen, and the “top 5 in Europe” ranking rests on the wider Green to Grey network’s data.
- 5

Verdict: largely supported (moderate confidence). The headline figure stands up and is, if anything, low. The farmland share is not shown, and the headline’s “over” 830,000 sits oddly with the text’s “nearly”.



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4 What the data show	8 Limitations

1 The claim and what we could verify

Amphora Media published the investigation on 11 September 2026 [1] with a companion piece on agricultural land the next day [2], in collaboration with Arena for Journalism in Europe and supported by The Malta Environment Foundation. We read both in full.

What was said	Where	Access
“nearly 830,000 square metres of nature and cropland have been built up in Malta between 2018 and 2023” (headline: “over 830,000”)	Amphora [1]	Read in full
Total land loss 0.26%, “among the top 5 countries”; “a conservative estimate”	Amphora [1]	Read in full
EEA: 920,000 m² (2012–2018) and 190,000 m² (2006–2012); “at least 1.94 square kilometres” since 2006	Amphora [1]	2012–18 checked [7]
“in nearly 95% of the take-up, farmland rather than natural area was affected”	Amphora [2]	Read in full

2 Method

Questions. Is the 830,000 m² figure plausible, are the comparisons right, and was the land mostly farmland?

Evidence. Amphora’s method [1] uses Google and WRI’s Dynamic World [3]. For an independent estimate we used a different model, Impact Observatory and Esri’s 10 m annual land cover for 2017–2023 [4, 5], through Microsoft Planetary Computer. Island areas are from OpenStreetMap. Scripts and data: *tools/cc-019-report/* and *data/cc-019/*.

Grades. Satellite land cover is a direct, model-classified measurement (grade B). ♦ marks a source known second-hand.

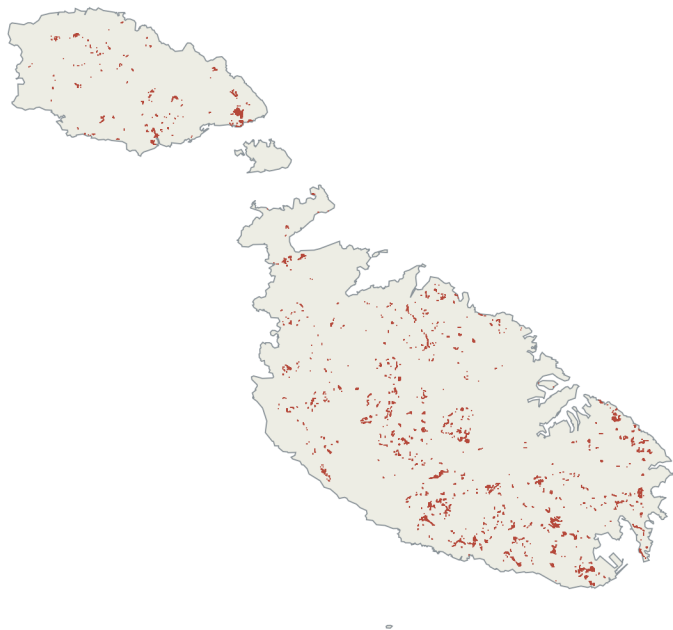
3 How the satellite data work

Both products classify every 10 m pixel with a deep-learning model trained on Sentinel-2 imagery [3, 4]. Single-year maps are noisy: in the Impact Observatory data Malta’s built-up area changes by 3 to 27 km² from one year to the next; the largest swing is over thirty times the 0.83 km² being measured. Comparing the 2018 and 2023 maps directly gives 24 km² of newly built-up pixels, or 18.9 km² net of pixels classed the other way, which is not credible. We therefore counted only pixels that were not built up in each of 2017, 2018 and 2019 and built up in each of 2021, 2022 and 2023, and subtracted the reverse change as a gauge of noise. This rule only detects land first mapped as built in 2020 or 2021; a looser two-year rule (not built in 2017–18, built in 2022–23) covers land first mapped as built in 2019–2022. Amphora handled the same problem by checking each flagged area by eye, which removes false alarms but can miss developments, as it notes.

4

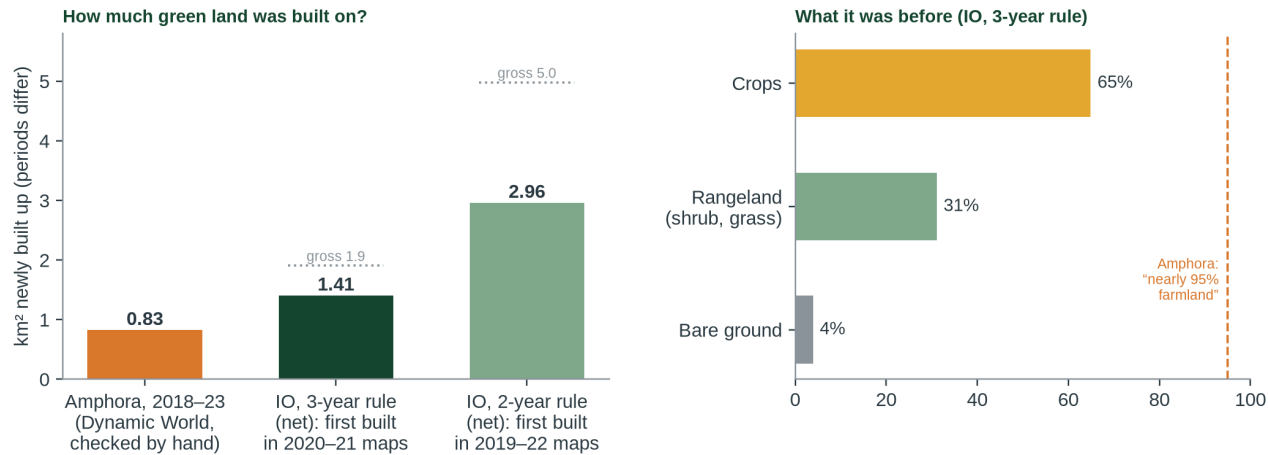
What the data show

Land that became consistently built up, 2017–19 to 2021–23 (red, 10 m pixels)



Source: Impact Observatory / Esri 10 m annual land cover v2 via Microsoft Planetary Computer; coastline © OpenStreetMap contributors. Pixels not built in 2017–2019 and built in 2021–2023.

Figure 1. Land that became consistently built up between 2017–19 and 2021–23 in the Impact Observatory data. The changes are small and scattered, as Amphora describes.



Net = new built-up minus the reverse change under the same rule (a gauge of classification noise). Dotted lines: gross change.

Figure 2. Left: Amphora’s figure (2018–2023) against two independent estimates, which cover land first mapped as built in 2020–21 and in 2019–22. Right: what the newly built land was before.

Check	Result	Amphora	Grade
830,000 m² as a share of Malta’s land (314 km²)	0.264%	0.26%	C
In FIFA pitches; vs Comino; vs Manoel Island	116; 0.24–0.30; 2.7	116; roughly a quarter; two	C
EEA 2006–12 + 2012–18 + Amphora 2018–23	1.94 km²	at least 1.94 km²	C
EEA land take 2012–18, read from chart [7]	0.91–0.94 km²; highest of 39	0.92 km²	C
New built-up, IO, 3-year rule (net; first built 2020–21)	1.41 km² (gross 1.91)	0.83 km² (2018–23)	B

Check	Result	Amphora	Grade
New built-up, IO, 2-year rule (net; first built 2019–22)	2.96 km² (gross 4.98)	–	B
Previously cropland / shrub and grass / bare	65% / 31% / 4%	nearly 95% farmland	B

All values in data/cc-019/checks.csv. Comino: 0.30 of its 2.8 km² OpenStreetMap outline, 0.24 of the 3.5 km² often quoted.

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Where the evidence points different ways

Q1 Is 830,000 m² right?		PLAUSIBLE, PROBABLY LOW
EVIDENCE FOR THE CLAIM'S IMPRESSION An independent model finds 1.4 to 3.0 km² net under strict rules, over windows shorter than Amphora's five years: more than Amphora's figure, which it calls conservative.	EVIDENCE AGAINST The independent estimates include classification noise that Amphora removed by hand; the true figure could be nearer Amphora's.	
WHY THEY DIFFER, AND WHAT IT MEANS HERE For this claim: the scale is right, and the figure is unlikely to be an overstatement.		
Q2 Was 95% of it farmland?		NOT SHOWN
EVIDENCE FOR THE CLAIM'S IMPRESSION Two thirds of the new built-up land was cropland in the independent data, and much Maltese shrubland is abandoned farmland.	EVIDENCE AGAINST A third was classed as shrub or grassland, which includes semi-natural garrigue. Amphora gives no basis or period for the 95%.	
WHY THEY DIFFER, AND WHAT IT MEANS HERE For this claim: “mostly farmland” is supported; “nearly 95%” is not reproduced.		
Q3 Over or nearly 830,000?		INCONSISTENT WORDING
EVIDENCE FOR THE CLAIM'S IMPRESSION The text says “nearly”, the headline “over”.	EVIDENCE AGAINST A headline should not round up a figure the text rounds down.	
WHY THEY DIFFER, AND WHAT IT MEANS HERE For this claim: a minor point that does not change the finding.		

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Testing the claim

Sub-claim	What the evidence shows	Rating
A. About 830,000 m ² of green land built up 2018–2023	Independent estimate 1.4–3.0 km ² net, over shorter windows (land first built 2019–22); Amphora's figure plausible and conservative.	SUPPORTED
B. 0.26% of land; comparisons (pitches, Comino, Manoel)	Arithmetic checks out (Comino: 0.24 of the 3.5 km ² often quoted, 0.30 of the OSM outline; Manoel: 2.7, not two).	SUPPORTED
C. Nearly 95% of take-up was farmland	Independent data: 65% cropland, 31% shrub or grass.	NOT SHOWN
D. EEA figures and the 1.94 km ² total	Sum correct; EEA chart gives 0.91–0.94 km ² for 2012–18 [7]; 2006–12 not seen.	SUPPORTED

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Verdict and requests for evidence

VERDICT

LARGELY SUPPORTED

The headline figure stands up and is probably low; the 95% farmland share is not shown. Confidence: moderate.

Why. (1) The figure is consistent with an independent satellite estimate, which finds as much or more. (2) The arithmetic and comparisons are right. (3) The farmland share and the EEA figures could not be verified, and the headline rounds up a number the text rounds down. These are caveats on details, not on the substance.

Evidence we are asking for

- ☐ Amphora's polygons (the map layer) and the basis and period of the 95% farmland figure.
- ☐ The EEA land-take value for Malta, 2006–2012.

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Limitations

- Land-cover models misclassify pixels; our rules reduce but do not remove this.
- The persistence rules only see land first mapped as built in 2020–21 (3-year rule) or 2019–22 (2-year rule), not the whole of 2018–2023.
- Built-up land includes roads, car parks and quarries as well as buildings; so does Amphora's.
- Neither estimate says whether development was inside or outside the development zone.
- Class names differ between the two models; 'rangeland' in IO is not the same as 'natural'.

References

- [1] Amphora Media (11 Sep 2026). Green To Grey: Malta lost over 830,000 square metres of land to development in five years. <https://www.amphora.media/2026/09/green-to-grey-malta-land-loss-development-construction>
- [2] Amphora Media (12 Sep 2026). Green to Grey: when development creeps up on agricultural land. <https://www.amphora.media/2026/09/green-to-grey-development-agricutlural-land>
- [3] Brown C.F. et al. (2022). Dynamic World, near real-time global 10 m land use land cover mapping. *Scientific Data* 9:251. doi:10.1038/s41597-022-01307-4. <https://doi.org/10.1038/s41597-022-01307-4>

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- [4] Karra K. et al. (2021). Global land use / land cover with Sentinel 2 and deep learning. *IGARSS 2021*, 4704–4707. doi:10.1109/IGARSS47720.2021.9553499. <https://doi.org/10.1109/IGARSS47720.2021.9553499>
- [5] Impact Observatory and Esri. 10 m Annual Land Use Land Cover (9-class) V2, 2017–2023; Microsoft Planetary Computer; retrieved 4 Oct 2026. <https://planetarycomputer.microsoft.com/dataset/io-lulc-annual-v02>
- [6] MiŻien. Data and calculations: data/cc-019/; tools/cc-019-report/.
- [7] European Environment Agency (2019, modified 2024). Country comparison: land take and land recultivation in EEA39 in the period 2012–2018 (in proportion of country area). Value read from the chart. <https://www.eea.europa.eu/en/analysis/maps-and-charts/country-comparison-land-take-and>

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient, or direct measurement · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** before wider circulation and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	4 Oct 2026	First issue. Right of reply to Amphora Media not yet sent.
1.1	5 Oct 2026	Corrections. (1) Periods of the independent estimates: “2018–2023” → land first mapped as built in 2020–21 (3-year rule) and 2019–22 (2-year rule), which is what the rules can detect (TL;DR, tile, sections 3, 4, 5 and 6, limitations, Figure 2 labels and caption). (2) Comino: “about 0.3 of Comino, as stated” → Amphora says “roughly a quarter”; 830,000 m² is 0.24 of the 3.5 km² often quoted and 0.30 of the 2.8 km² OpenStreetMap outline. (3) Simple comparison: “would give 24 km²” → 24 km² of newly built-up pixels, 18.9 km² net (now saved in <i>data/cc-019/io_lulc_change.csv</i>). (4) Year-to-year noise: “more than 15 km², twenty times the change” → 3 to 27 km², the largest over thirty times 0.83 km²; flyer “15 km² a year” → “up to 27 km² from year to year”. (5) The TL;DR now says the two estimates are not like for like. Verdict and confidence unchanged.